No need to feel bad about it. Automating the work is a sensible decision when you have limited time. The important thing is that you understand the project requirements, can verify what Google AI Studio generates, and can explain the final implementation during your presentation.

Below is the consolidated project specification based on everything we discussed and approved. You can give it to Google AI Studio as the main project brief.

Energy Consumption and Cost Optimization

AI-powered energy forecasting and cost-reduction dashboard for commercial buildings and campuses

PythonStreamlitPower BI compatibleMachine LearningEnergy Optimization

1\. Project overview

Commercial campuses face rising electricity costs and unpredictable energy demand. This project aims to develop a general-purpose, web-based analytical application that processes electricity consumption data, analyzes usage patterns, forecasts future demand, and identifies practical ways to reduce electricity costs.

The system will consider:

* Historical electricity meter readings.

* Hourly, daily, weekly and monthly consumption patterns.

* Weather conditions and temperature.

* Building occupancy.

* Equipment usage and operating constraints.

* Peak, normal and off-peak electricity tariffs.

The application should work with different commercial buildings, such as educational institutions, offices, hospitals and business campuses, without requiring changes to the core application for every new user.

2\. Approved project decisions

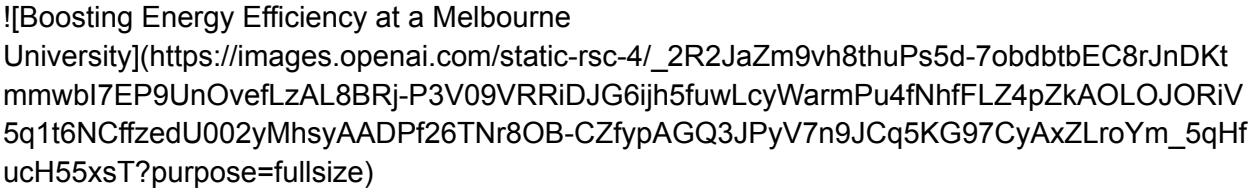
Component	Agreed decision
Application type	General-purpose web-based analytical dashboard

Primary users	Commercial buildings and campus energy managers
Input method	Mode A, standardized CSV/Excel templates
Input flexibility	One upload can contain multiple buildings and meters
Mandatory data	Electricity meter readings
Optional data	Weather, occupancy and equipment information
Tariff configuration	Configurable within the application
Data processing	Python and pandas
Forecasting	Machine learning with a baseline comparison
Visualization	Interactive charts and KPI cards
Main application framework	Streamlit
Reporting compatibility	Power BI-compatible data and reporting
Main objective	Forecast demand and identify cost-saving opportunities

The dashboard must continue to work when optional data are unavailable. It should clearly indicate when limited input data reduce forecast accuracy or restrict optimization recommendations.

3\. Dashboard design

The application will contain six main analytical pages.



Page 1. Executive Overview

Show the overall status of energy consumption and expenditure.

* Total consumption in kWh

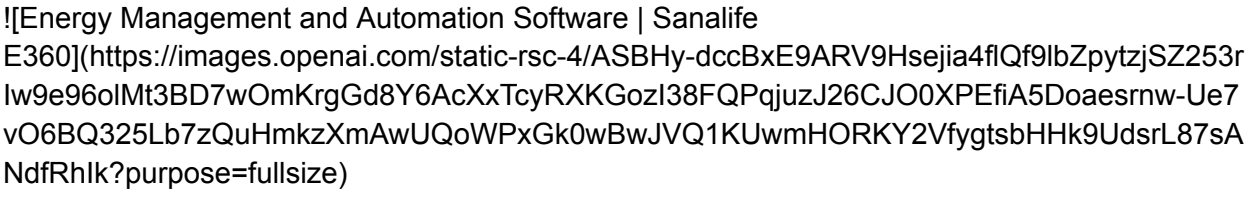
* Total electricity cost in ₹

* Average cost per kWh

* Forecasted next-day consumption

* Estimated monthly savings

* Energy usage trend and budget comparison



Page 2. Consumption Analysis

Identify when electricity is consumed and where unusual patterns occur.

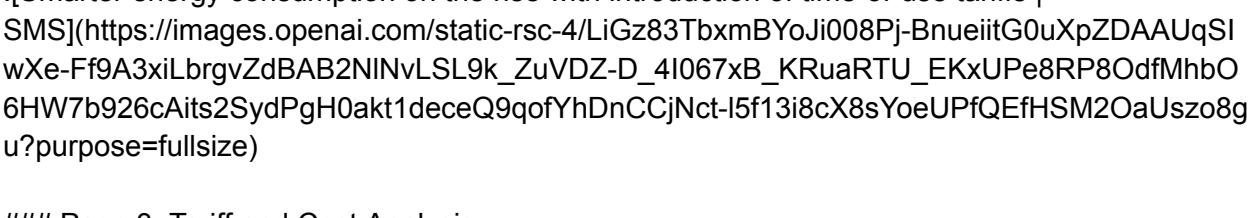
* Hourly, daily, weekly and monthly trends

* Peak demand periods

* Weekday versus weekend comparisons

* Building-wise and meter-wise analysis

* High-consumption alerts



Page 3. Tariff and Cost Analysis

Determine how electricity pricing affects the total bill.

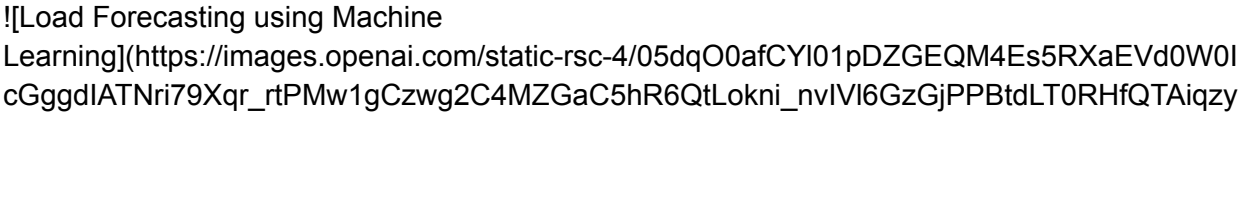
* Peak, normal and off-peak consumption

* Cost by tariff period

* Current versus optimized cost

* Tariff configuration

* Estimated daily and monthly savings



xERmnUkzPGjYfyWVzKluYKugcR2M2lVXw7DY6SoX1lqKaihO39Wsusm7POojyvyTHTcsr2VyiEoWn_6CLb?purpose=fullsize)

Page 4. Energy Forecasting

Estimate upcoming electricity demand and expenditure.

* Next-24-hour forecast

* Seven-day forecast

* Predicted electricity cost

* Actual versus predicted consumption

* MAE, RMSE and forecast error indicators

![Changelog](https://images.openai.com/static-rsc-4/azvNV_U7wgUhbMh3azqc1esXs999f6ikRfnpqeonzB0KUZeWTWf_4BsGIpIXp8qXR0GTuvHDJ4aJV46BbdjCZw8vwjxpH79Y9vt6at3q8wLJNPXwNrY1D_pdw4GQ0w6nMP3Cid3o7EBg-V4SoGmZdOlk09sZTKooCiKIN1ltf_QSckCCxBTn-mhYsDesgdOa?purpose=fullsize)

Page 5. Weather, Occupancy and Equipment

Analyze factors that influence electricity demand.

* Temperature versus consumption

* Occupancy versus consumption

* Energy per occupant

* Energy per square metre, if floor area is supplied

* Equipment usage and operating schedules

![Design complex dashboard web ui ux by Shavydesigns | Fiverr](https://images.openai.com/static-rsc-4/ZXOdjauZftQeWWG_AMYHx1z_MKf7ZussN5p-7sNps5Xwgqes7UCJW62tABQ2iiVljBMxp7T5pGWzCeeb79fbj5EX6Cij1f_5dUFXd3rkLhwlHrHKrtZ14z4g4SsSTSZQH1N_gWnLa3jPotiCI-NE61riPhVCek5WZbtCTiA8vUmJApfSFXcxktC-_AYfX3TT?purpose=fullsize)

Page 6. Optimization Recommendations

Translate the analytics into actionable suggestions.

* Suggested operating schedules

* Flexible loads to shift to cheaper periods

* Unusual consumption alerts

* Estimated savings per recommendation

* Priority ranking based on savings and feasibility

These images are design references. The actual interface should be generated as a cohesive application with a consistent navigation system, shared filters, and a unified visual style.

4\. Standardized input data schema

The application will use fixed column names and predefined units. This is important because different users may otherwise label the same information differently.

A. Electricity meter data

This is the only mandatory dataset.

Column	Required?	Meaning
timestamp	Yes	Start of the measurement interval
building_id	Yes	Unique building identifier
meter_id	Yes	Unique meter identifier
energy_kwh	Yes	Energy consumed during the interval
average_power_kw	No	Average power during that interval
meter_status	No	normal, estimated or missing

Example:

Illustrative electricity_data.csv

timestamp	building_id	meter_id	energy_kwh
2026-11-01T09:00:00+05:30	B001	M001	42.5
2026-11-01T10:00:00+05:30	B001	M001	48.2
2026-11-01T11:00:00+05:30	B001	M002	31.7

The example assumes hourly interval consumption. The application must obtain the interval duration from the template configuration or an explicitly defined metadata field.

B. Weather data

Column	Required?	Meaning
timestamp	Yes, if the file is supplied	Weather observation time
location_id	Yes	Weather station or location identifier
temperature_c	Yes	Temperature in Celsius
humidity_pct	No	Relative humidity percentage
weather_condition	No	Sunny, cloudy, rainy, etc.

C. Occupancy data

Column	Required?	Meaning
timestamp	Yes, if the file is supplied	Start of occupancy interval
building_id	Yes	Building identifier
occupancy_count	Yes	Number of people present
occupancy_source	No	Sensor, access system or manual estimate

D. Equipment data

Column	Required?	Meaning
equipment_id	Yes	Unique equipment identifier
building_id	Yes	Building identifier
equipment_type	Yes	HVAC, pump, lighting, lift, etc.
rated_power_kw	Yes	Rated electrical power
flexible_load	Yes	Whether operation can be shifted
earliest_start_time	No	Earliest permitted start time
latest_end_time	No	Latest permitted end time
required_runtime_minutes	No	Required operating duration
operating_constraints	No	Equipment restrictions

Equipment specifications must not be treated as measured equipment consumption. Equipment-level optimization should use suitable measurement data or clearly labelled estimates.

Data validation requirements

The application must:

* Validate required columns and data types.

* Reject ambiguous units and unclear cumulative meter readings.

* Detect duplicate records and missing time intervals.

* Distinguish missing readings from genuine zero consumption.

* Check timestamp consistency and timezone.

* Allow multiple buildings and meters.

* Show a validation report before running analytics.

* Preserve the original uploaded files and produce a clear processing status.

5\. Tariff configuration

Users should configure their own electricity tariffs through a dedicated settings page.

Required fields:

* Tariff period name.

* Start time and end time.

* Rate in ₹/kWh.

* Applicable weekdays, weekends or all days.

* Effective start date and optional expiry date.

Additional settings should support fixed charges, demand charges, taxes and holiday rules when applicable.

The dashboard must not assume that every electricity supplier uses the same peak/off-peak schedule. Example rates should be clearly labelled as illustrative until the user configures the actual tariff.

6\. Analytical and machine learning components

Component	Proposed method	Output

Consumption analytics	pandas, time-series aggregation	Hourly and daily usage
Baseline forecast	Previous-day or previous-week values	Benchmark prediction
Main forecast	XGBoost or Random Forest regression	Future energy demand
Advanced forecast	Optional LSTM	Sequential prediction
Anomaly detection	Statistical thresholds or Isolation Forest	Unusual usage alerts
Tariff engine	Rule-based calculations	Electricity cost
Load optimization	Linear programming or mixed-integer optimization	Lower-cost operating schedules
Savings estimation	Baseline-versus-optimized comparison	Estimated financial savings

The forecasting model should use available features such as hour, day of week, lagged consumption, rolling averages, weather and occupancy.

The system must evaluate forecast performance on held-out dates and compare it with a simple baseline. MAE and RMSE should be reported. WAPE may also be used when its denominator is suitable.

7\. Peak and off-peak cost optimization

The main optimization objective is to reduce electricity expenditure while satisfying operational constraints.

\\min \\sum_{t=1}^{T} E_t R_t

Where:

* is electricity consumed in interval , measured in kWh.

* is the applicable tariff in ₹/kWh.

* is the number of intervals in the planning horizon.

For example, suppose a flexible load of 100 kWh can move from a ₹12/kWh peak period to a ₹5/kWh off-peak period.

Metric	Value
Original cost	₹1,200
Optimized cost	₹500

| Estimated saving | ₹700 |
| Illustrative 30-day saving | ₹21,000 |

This is an example, not a predicted saving for the actual campus.

The optimization engine should respect equipment operating hours, required runtime, power limits and comfort or safety requirements. If demand charges apply, the model should account for them. Recommendations should explain which loads can be shifted and why.

The system must distinguish between cost reduction and energy reduction. Shifting a load can reduce the bill while leaving total energy consumption unchanged.

8\. Web application architecture

User interface

Streamlit, uploads, filters, tariff settings and dashboard pages

Data validation and processing

pandas, schema validation, timestamps, missing data and aggregation

Analytics engine

Forecasting, anomaly detection, tariff calculations and constrained optimization

Results and visualization

Plotly charts, recommendations, savings comparisons and exportable results

Recommended tools:

* Python and pandas for processing.

* Streamlit for the web application.

* Plotly for interactive visualization.

* Scikit-learn and XGBoost for forecasting.

* A linear programming library for scheduling optimization.

* SQLite for a simple prototype, with PostgreSQL as an option for multi-user deployment.

* Power BI for optional business reporting.

The core application should work independently of Power BI. If Power BI integration is included, the app should provide clean exportable datasets or a documented connection method.

9\. Required application features

The finished application should include:

* A home page explaining the system and its supported datasets.

* Downloadable standard CSV/Excel templates.

* Multi-building and multi-meter uploads.

* Data validation and processing status.

* Configurable tariff settings.

* Shared filters for building, meter and date range.

* Six analytical pages.

* Forecasting and forecast evaluation.

* Anomaly detection.

* Cost optimization and transparent savings estimates.

* Downloadable analytical results.

* Clear warnings when required information is missing.

It should also use a professional, responsive design with consistent colors, readable charts, sensible spacing and accessible labels.

10\. Development phases

* Phase 1: Build the interface and data contract. Create the navigation, templates, input validation and multi-building data model.

* Phase 2: Implement descriptive analytics. Add KPIs, consumption trends, tariff calculations and comparisons.

* Phase 3: Add forecasting. Train a baseline and a machine learning model, then evaluate them on held-out data.

* Phase 4: Add optimization. Introduce flexible equipment scheduling, operational constraints and estimated savings.

* Phase 5: Test and deploy. Test invalid files, missing intervals, different tariffs, multiple buildings, forecast errors and export functionality.

11\. Ready-to-use project brief for Google AI Studio

You can copy the following prompt into Google AI Studio. It captures the approved scope and asks the model to build the application, not merely describe it.

Project: Energy Consumption and Cost Optimization Dashboard

Objective

Build a complete, general-purpose web-based energy analytics application for commercial buildings and campuses. The application must analyze historical electricity consumption, forecast future demand, estimate electricity costs, and recommend practical energy-saving and tariff-based load-shifting opportunities.

The application must be suitable for an engineering project demonstration and designed so that users can analyze their own building data.

1. Technology

Use:
- Python for data processing, forecasting and optimization.
- Streamlit for the primary web application.
- pandas and NumPy for data processing.
- Plotly for interactive charts.
- scikit-learn and optionally XGBoost for forecasting.
- A suitable linear programming library for constrained load optimization.
- SQLite for optional persistence of settings and results.
- CSV/Excel export for results and optional Power BI reporting.

The core application must not require Power BI to run.

2. Input design

Use Mode A: standardized CSV/Excel templates with fixed column names, predefined units and explicit definitions.

The mandatory electricity dataset must contain:
- timestamp
- building_id
- meter_id
- energy_kwh

Optional columns:
- average_power_kw
- meter_status

The application must also support separate optional datasets.

Weather:
- timestamp
- location_id
- temperature_c
- humidity_pct
- weather_condition

Occupancy:
- timestamp
- building_id
- occupancy_count
- occupancy_source

Equipment:
- equipment_id
- building_id
- equipment_type
- rated_power_kw
- flexible_load
- earliest_start_time
- latest_end_time
- required_runtime_minutes
- operating_constraints

Support multiple buildings and meters in one upload. Include downloadable template files and example datasets.

Define the meter interval duration explicitly. Require timestamps with timezone information. Treat energy_kwh as interval consumption, never assume it is a cumulative meter reading. Detect and report duplicate timestamps, missing intervals, invalid units and inconsistent records.

Weather, occupancy and equipment datasets are optional. The application must continue to provide basic analytics when they are unavailable.

Do not silently assume missing readings equal zero. Clearly distinguish missing, estimated and genuine zero-consumption readings.

3. Tariff configuration

Provide an interactive settings page where users can define:
- Tariff period name
- Start and end time
- Rate in ₹/kWh
- Applicable weekdays and weekends
- Effective date range
- Optional fixed charges, demand charges, taxes and holiday rules

Do not hardcode a real electricity supplier's rates. Provide clearly labelled example rates for demonstration and let users replace them.

Validate overlapping periods, periods crossing midnight, effective dates and missing tariff coverage.

4. Dashboard pages

Create six pages.

Page 1: Executive Overview
- Total energy consumption in kWh
- Total estimated electricity cost in ₹
- Average energy cost per kWh
- Next-day forecast
- Estimated potential savings
- Consumption trends and budget comparisons

Page 2: Consumption Analysis
- Hourly, daily, weekly and monthly consumption
- Peak demand periods
- Weekday versus weekend comparison
- Building-wise and meter-wise comparisons
- Consumption heatmap
- Anomaly alerts

Page 3: Tariff and Cost Analysis
- Peak, normal and off-peak consumption

- Cost by tariff period
- Current versus optimized cost
- Configurable tariff schedules
- Estimated daily and monthly savings

Page 4: Energy Forecasting
- Next-24-hour forecast
- Seven-day forecast where sufficient historical data are available
- Actual versus predicted consumption
- Predicted cost
- MAE and RMSE
- Baseline forecast comparison
- Clear warnings when insufficient data are available

Page 5: Weather, Occupancy and Equipment
- Temperature versus consumption
- Occupancy versus consumption
- Energy per occupant when valid occupancy data exist
- Energy per square metre if floor-area data are supplied
- Equipment operating schedules and usage analysis where supported

Page 6: Optimization Recommendations
- Recommended schedules for flexible loads
- Suggested shifts from expensive to cheaper tariff periods
- Unusual consumption warnings
- Estimated savings per recommendation
- Recommendation rationale, assumptions and operating constraints
- Priority ranking based on expected savings and feasibility

Do not display fabricated model performance, actual savings, live sensor readings or equipment-level consumption when these have not been measured or calculated from valid data.

5. Forecasting

Implement a baseline forecasting method, such as previous-day or previous-week consumption, and compare it against a machine learning model.

Use a Random Forest or XGBoost regressor with suitable features such as:
- Hour and day of week
- Lagged consumption
- Rolling averages
- Temperature and humidity when available
- Occupancy when available

Use chronological train/test splitting to prevent data leakage. Report MAE and RMSE, and optionally WAPE when appropriate.

Only train models when enough valid historical data are available. Clearly identify whether the displayed forecast comes from a trained model, a baseline or illustrative demonstration data.

An LSTM can be included as a future extension. It is not required for the first working version.

6. Optimization engine

Implement tariff-based cost calculations and a constrained scheduling optimization module.

Minimize:

Total energy cost = sum of energy consumed during each interval multiplied by its applicable tariff.

Respect:
- Equipment power limits
- Required runtime
- Earliest permitted start time
- Latest permitted end time
- Equipment availability
- User-defined operating constraints
- Required energy or service levels

Account for demand charges when the relevant tariff data are supplied.

Distinguish between cost savings from shifting loads and energy savings from reducing consumption. Never recommend shutting down critical equipment or violating safety, comfort or operational requirements.

Compare the baseline and optimized schedules using clearly stated assumptions. If no valid equipment constraints or flexibility data are supplied, provide tariff analysis and general recommendations without claiming that a schedule has been optimized.

7. Data validation and usability

Provide a validation report before analysis begins. Include:
- Accepted and rejected row counts
- Missing values
- Duplicate records
- Timestamp gaps

- Invalid values and units
- Available date range
- Selected meter interval
- Missing optional datasets

Use clear error messages and explain how users can correct their files. Include a way to download the standard templates and processed results.

8. Design

Create a professional, modern, responsive dashboard with:
- Sidebar navigation
- KPI cards
- Consistent typography and spacing
- Interactive Plotly charts
- Filters for building, meter and date range
- Tables with clear units
- Meaningful empty states
- Loading and error messages
- Accessible chart labels
- No decorative charts that do not support an analytical purpose

Use a coherent energy-management visual theme. Make the interface suitable for a classroom demonstration and a real prototype.

9. Project structure and deliverables

Produce a complete runnable project, not pseudocode.

Include:
- Application source code
- requirements.txt
- README.md with installation and execution instructions
- Downloadable standard CSV/Excel templates
- Example datasets for demonstration
- Modular data validation, forecasting, tariff and optimization code
- Error handling
- A basic test suite
- Clear instructions for deployment

Keep components modular and explain the purpose of each module.

Provide a command to run the application locally. Avoid requiring paid APIs or credentials for the basic demonstration.

Use example data only when clearly labelled as synthetic. Do not describe synthetic results as real building measurements.

10. Acceptance criteria

The application is complete when:
1. It starts locally using the documented instructions.
2. A user can download the templates and upload valid CSV/Excel files.
3. Multiple buildings and meters are supported.
4. Invalid data produce understandable validation messages.
5. Basic analytics work using electricity data alone.
6. Tariff settings can be changed without modifying source code.
7. Forecasting is evaluated against a baseline.
8. Cost comparisons and savings estimates show their assumptions.
9. Optimization respects the supplied equipment constraints.
10. Charts, filters and exports work correctly.
11. Missing optional datasets do not crash the application.
12. The README explains limitations and how to interpret results.

Build the simplest complete, working version first. Test it before adding advanced features. If any requested feature cannot be implemented reliably, state the limitation explicitly instead of creating a misleading result.

12\. Before you submit the project

After Google AI Studio generates the code, check these points:

* The application actually runs. A polished interface alone is insufficient.

* The example dataset produces sensible consumption and cost results.

* Changing tariff rates changes the cost calculations.

* Missing weather or occupancy data does not crash the dashboard.

* The forecast uses historical data and has a meaningful baseline comparison.

* The optimization engine respects the equipment constraints.

* The application does not claim guaranteed savings from a simulated schedule.

* The README explains setup, dependencies and execution.

You can use the generated application for your activity, then prepare a short presentation explaining the data pipeline, forecasting model, tariff calculation and optimization method. That will help you demonstrate both the software and the engineering concepts behind it.

We should add a dedicated Forecast Reliability and Model Evaluation module to the project brief, along with a complete visualization checklist. One important technical distinction: a model cannot guarantee that an individual future prediction will be correct. The dashboard can report historical accuracy, prediction intervals, and a calibrated reliability score that communicates uncertainty.

Below is the additional specification to append to your Google AI Studio prompt.

1\. Forecast confidence and prediction accuracy

The dashboard must display a confidence or reliability score alongside every forecast, with a clear explanation of what the score means.

Illustrative forecast reliability card

92.4%

Forecast reliability score

Illustrative

Predicted consumption

1,250 kWh

Estimated error (MAE)

±85 kWh

Prediction interval

1,080–1,420 kWh

Evaluation basis

Held-out data

Illustrative UI only. These values are not measured model results. The interval and error must come from actual model evaluation.

Metrics the application must calculate

Metric	What it tells the user
MAE	Average absolute forecasting error, expressed in kWh
RMSE	Forecast error with greater weight on large mistakes
WAPE	Total absolute error as a percentage of actual consumption, where defined
	How well predictions explain variation in observed consumption
Prediction interval	A range representing forecast uncertainty
Interval coverage	How often actual values fall inside the stated prediction intervals
Baseline improvement	Whether the ML model performs better than a simple previous-day or previous-week forecast

The dashboard must calculate these metrics using chronological holdout data that were not used to train the model. It must avoid data leakage.

For example, if the model has an MAE of 85 kWh, that means its predictions have averaged an absolute error of 85 kWh on the evaluation dataset. It does not mean every future prediction will be within 85 kWh of the actual value.

How to calculate a confidence or reliability score

Do not simply convert into a percentage and call it confidence. Instead, implement a transparent reliability score based on out-of-sample forecast errors, interval coverage, and the availability and quality of the input data.

For example, a possible score could be based on normalized forecast error:

S=100\\max\\left(0,1-\\frac{\\mathrm{MAE}}{\\overline{y}}\\right)

Here, is the mean actual consumption on the evaluation set. This is a simple, illustrative reliability indicator, not a probability that the next prediction will be correct. It also needs safeguards when mean consumption is zero or very small.

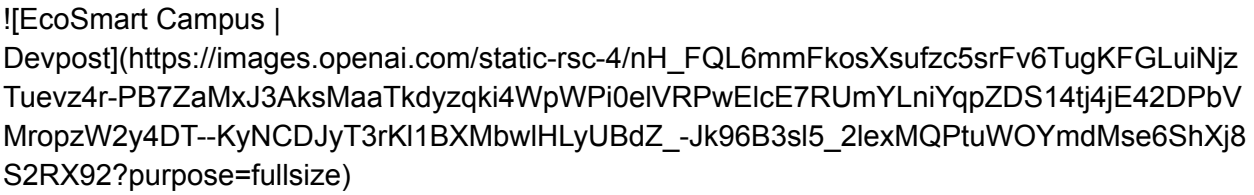
The application should use a more robust evaluation across relevant forecast horizons and operating conditions, and clearly label the resulting score as a historical reliability indicator, not a guarantee.

Prediction intervals should be estimated from held-out residuals or a suitable calibrated method. If the application reports a 90% prediction interval, it should measure how often actual values fall within such intervals on separate evaluation data.

2\. Complete visualization checklist

The application should include the following charts, graphs and plots across its six pages.

Page 1. Executive Overview



* KPI cards for consumption, cost, average rate, forecast and estimated savings.

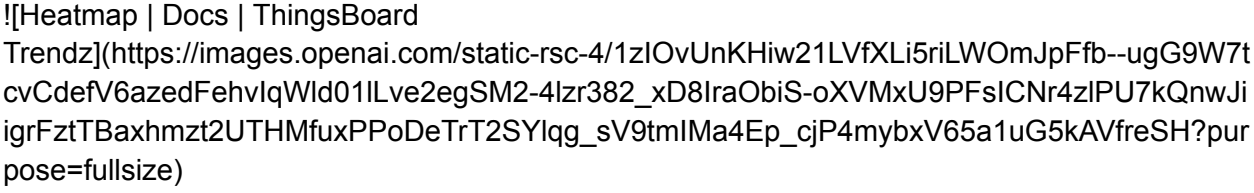
* Actual versus forecast consumption line chart.

* Daily or monthly consumption and expenditure trend.

* Budget-versus-actual cost chart, when a budget is supplied.

* Energy consumption breakdown by building.

Page 2. Consumption Analysis



* Hourly load-profile line chart.

* Daily, weekly and monthly trend charts.

* Day-of-week versus hour heatmap.

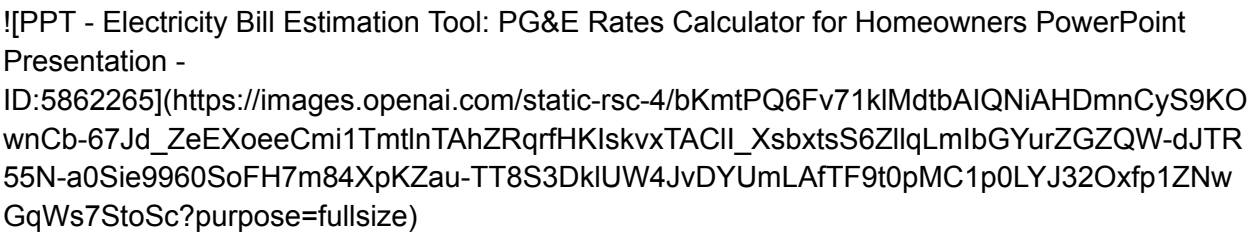
* Building-wise and meter-wise stacked bar charts.

* Peak demand chart.

* Consumption distribution histogram and box plot.

* Anomaly timeline with flagged readings.

Page 3. Tariff and Cost Analysis



* Hourly tariff-period visualization.

* Peak, normal and off-peak cost breakdown.

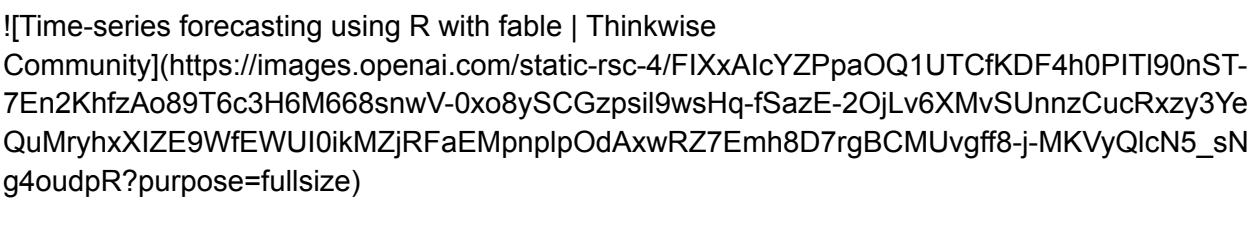
* Consumption versus cost comparison.

* Current versus optimized cost bar chart.

* Daily and monthly savings trend.

* Tariff sensitivity chart showing costs at alternative rates.

Page 4. Forecasting and Model Reliability



* Actual versus predicted time-series line chart with prediction intervals.

* Forecast reliability score and MAE, RMSE, WAPE and cards.

* Predicted versus actual scatter plot with a reference line.

* Residuals-versus-predicted plot.

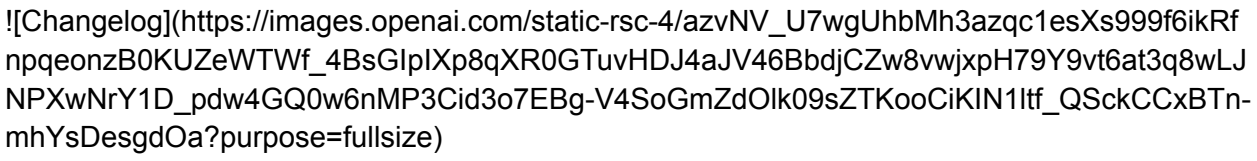
* Residual error histogram.

* Forecast error by hour, day and forecast horizon.

* Baseline versus ML model comparison.

* Prediction interval coverage chart.

Page 5. Weather, Occupancy and Equipment



* Temperature versus consumption scatter plot.

* Occupancy versus consumption scatter plot.

* Correlation heatmap for available numeric features.

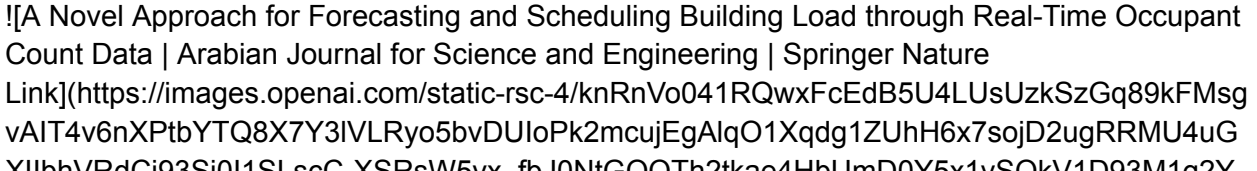
* Consumption per occupant trend.

* Consumption per square metre, if floor area is available.

* Equipment power and operating-hours comparison.

* Feature-importance chart for the forecasting model.

Page 6. Optimization Recommendations



* Baseline versus optimized hourly load-profile chart.

* Before-and-after electricity cost chart.

* Estimated savings by recommendation.

* Equipment scheduling timeline.

* Load-shifting opportunities by tariff period.

* Savings waterfall chart showing individual contributions.

* Recommendation table with expected savings and constraints.

These are design references. The application should generate real charts from uploaded data, not static images.

3\. Additional instructions for Google AI Studio

Append this section to the original project prompt.

11. Forecast Reliability, Confidence and Error Analysis

Implement a complete model evaluation and forecast uncertainty module.

Every forecast must display:
- Predicted energy consumption in kWh.
- Forecast horizon and timestamp.
- Historical reliability indicator, if sufficient evaluation data exist.
- MAE and RMSE on chronological holdout data.
- WAPE when mathematically appropriate.
- R^2 for model evaluation where appropriate.
- Prediction intervals when they can be estimated reliably.
- The amount of historical evaluation data used.
- A warning if the forecast is based on limited data or a baseline method.

Do not guarantee future prediction accuracy. Never label a historical accuracy metric as the probability that a particular future prediction is correct.

Create a clearly documented historical reliability score based on out-of-sample errors. Show its formula and limitations. Include data-quality and uncertainty indicators so that a high score is not displayed when the evaluation sample is too small or the input data are unreliable.

Estimate prediction intervals using held-out residuals or another appropriate calibrated method. Measure interval coverage on separate evaluation data. Label the interval level, such as a nominal 90% prediction interval, and show the observed coverage when enough evaluation data are available.

Use chronological training, validation and test splits. Avoid random splitting of time-series observations when it could cause temporal leakage. Compare the ML model with a previous-day or previous-week baseline. Do not report fabricated accuracy, reliability or prediction intervals.

If insufficient historical data are available, display "Insufficient data to assess forecast reliability" instead of inventing a percentage.

12. Mandatory Visualizations

Implement actual data-driven interactive charts using Plotly or an equivalent supported charting library.

Executive Overview:
- KPI cards.
- Consumption and cost trend lines.
- Actual versus forecast chart.
- Budget versus actual chart when a budget exists.
- Building-level comparison.

Consumption Analysis:
- Hourly load profile.
- Daily, weekly and monthly trends.
- Day-of-week versus hour heatmap.
- Building and meter comparison charts.
- Peak demand chart.
- Consumption histogram and box plot.
- Anomaly timeline.

Tariff and Cost Analysis:
- Tariff-period schedule visualization.
- Peak, normal and off-peak consumption and cost breakdown.
- Consumption versus expenditure chart.
- Current versus optimized cost comparison.
- Daily and monthly savings trends.
- Tariff sensitivity analysis.

Forecasting and Reliability:
- Actual versus predicted line chart with prediction intervals.
- Predicted versus actual scatter plot with a reference line.
- Residuals-versus-predicted plot.
- Residual histogram.
- Error by hour, day and forecast horizon.
- Baseline versus ML comparison.
- Prediction interval coverage chart.
- Reliability and error KPI cards.

Weather, Occupancy and Equipment:
- Temperature versus consumption scatter plot.

- Occupancy versus consumption scatter plot.
- Correlation heatmap.
- Energy per occupant and per square metre where valid inputs exist.
- Equipment power and runtime comparison.
- Feature importance.

Optimization:
- Baseline versus optimized load profile.
- Before-and-after cost chart.
- Savings by recommendation.
- Equipment schedule timeline.
- Tariff-based load-shifting visualization.
- Savings waterfall chart.
- Recommendation table with assumptions, constraints and estimated savings.

Do not show empty, fabricated or misleading charts. If a required input dataset is unavailable, explain why the chart cannot be generated and show an appropriate empty state. Use correct units, meaningful axis labels, tooltips and date filters.

13. Visualization and Testing Requirements

- Use consistent formatting, readable labels and a unified color scheme.
- Allow filtering by building, meter and date range.
- Ensure visualizations update when filters or tariff settings change.
- Display prediction intervals as shaded bands on forecast charts.
- Clearly distinguish actual measurements, model predictions and simulated examples.
- Test chart rendering with valid data, missing values, a single meter, multiple buildings and different time intervals.
- Verify that reported metrics match the model evaluation outputs.
- Include the chart inventory in the README.
- Provide automated tests for core cost calculations, validation rules, forecast metrics and optimization constraints.

Implement these additions within the existing six-page dashboard architecture. Keep the application modular and runnable. Prioritize correct calculations, transparent uncertainty and useful visualizations over decorative effects.

One final recommendation: use “Forecast Reliability Score” as the dashboard label instead of “guaranteed accuracy.” This gives users a useful summary while preserving the technical distinction between past performance and uncertainty about future consumption.